Motivation Letter: AI Agent & Tool-Use Engineer (RE2)

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Recruitment Panel
Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for AI Agent & Tool-Use Engineer (RE2), reference 187_26_AII_LM_RE2

Dear Members of the Recruitment Panel,

I am applying for the AI Agent & Tool-Use Engineer position because it sits at the intersection I have been building towards: reliable agentic workflows, well-defined tool interfaces, and evaluation that distinguishes model failures from failures in the surrounding system. I am completing an MSc in Modelling for Science and Engineering at the Universitat Autonoma de Barcelona, specialising in Data Science. Alongside my studies, I work on enterprise agent systems at Aily Labs and completed a JAE Intro ICU 2025 research scholarship at IIIA-CSIC.

At Aily Labs, I work on the engineering needed to turn language models into production analytical agents. My contributions have included prompt-defined MCP capabilities, an MCP server package for go-to-market workflows, and the transition towards task-scoped skills that control which instructions and tools are exposed. I have also worked with a semantic access layer in which curated SQL table descriptions encode business meaning and intended usage. This allows an SQL agent to query approved data through a bounded interface instead of inferring how to use a database from raw schema alone.

Evaluation closes that production loop. Package-owned question sets declare the expected tables, skills, tools, and answer criteria for each capability. Benchmark runs combine deterministic assertions, LLM-as-a-judge scoring, and trace inspection, with the resulting checks incorporated into CI/CD and promotion decisions. This makes failures attributable: the evidence can show whether a weak answer arose from model reasoning, tool selection, semantic context, routing, or runtime execution. It also provides a disciplined way to improve an agent through better contracts, context, and tools before assuming that fine-tuning is the only remedy.

At IIIA-CSIC, I applied the same approach to an embodied system. My MSc thesis integrates dialogue, symbolic knowledge, language-model planning, deterministic execution, and typed feedback in a ROS 2 architecture for a NAO robot. The system constrains planning through a reviewed skill registry, passes compact knowledge context to the language-model components, preserves execution lineage, and validates both simulated and physical skill adapters through structured traces. I designed the validation around dialogue, knowledge interaction, single and composite skills, and failures introduced at the beginning, middle, or end of execution. This work reinforced a practical lesson: fluent model output is not evidence of reliable tool use unless the capability exposure, world state, action result, and trace are also correct.

These lessons also motivated an unpublished proof of concept for a Universal Agentic Harness. It projects task-specific capabilities from a registry, applies deterministic reach and effect checks, and records append-only traces through a portable environment interface. The proof of concept has focused tests but is not yet live-integrated or externally validated. Future work will test the same contracts against the other agentic environments in my project portfolio. The objective is concrete: make capability exposure, permissions, and evidence requirements explicit enough that an agent can be evaluated consistently across different domains.

My strongest personal machine-learning project, iTrader, provides complementary model-side experience. Its implemented research environment uses a PPO/GAE solver, task-conditioned FiLM policy and value networks, validated task specifications, and a decomposed reward engine covering PnL, drawdown, turnover, and transaction-related effects. Runs preserve configuration, seed, code, and dataset lineage. Rule-based and local LLM proposer backends generate candidate curricula behind validation and deterministic controls. Planned work will evaluate grouped proposer candidates and GRPO-like optimisation once the downstream reward signal is sufficiently robust. This has given me direct experience with PyTorch, policy optimisation, reward shaping, experimental controls, and the failure modes of learning systems that optimise against an imperfect objective.

My hands-on experience is strongest in model-tool systems, PPO/GAE, reward design, and evaluation harnesses. I have less production experience with TRL, FastChat, NeMo-RL, DPO, GSPO, and CISPO, and I would treat these as a focused area of development. I would not, however, approach them without the necessary engineering foundation. My work already relies on typed task contracts, separate development and holdout evidence, reproducible runs, reward-hacking checks, and evaluation that can distinguish a better policy from a more permissive runtime.

The Language Modeling Team is therefore a strong fit for both what I can contribute immediately and where I want to deepen my research engineering. I can bring practical experience in MCP integration, function calling, task-oriented tool exposure, evaluation pipelines, trace analysis, and Python ML development. I am equally motivated to apply that experience to instruction data curation and post-training experiments on BSC infrastructure. I work fluently in English and Spanish, and I value the opportunity to contribute to open language technologies within a multidisciplinary research environment.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience in agent evaluation, model-tool integration, and reproducible ML experimentation could support the team.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Roger Montane**, Mid Data Scientist, Aily Labs

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2. **Dr. Balogh Miklos**, Professor, BME

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